Exercises: Cones of K3 Surfaces

Hajime Nakahashi

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Throughout, we work over $\mathbb{C}$. For each of the following K3 surfaces X, determine

$$\boxed{\rho(X), \qquad \mathrm{Amp}(X), \qquad \mathrm{Nef}(X), \qquad \mathrm{NE}(X).}$$

Describe these cones explicitly in $\mathrm{NS}(X)_{\mathbb{R}}$, and determine whether each cone is closed and whether it is rational polyhedral.

Remark. Here $\mathrm{NE}(X)$ denotes the effective cone of curves.

Unless otherwise stated, very general means outside a countable union of proper closed algebraic subsets of the relevant parameter space. Standard results on generic Picard groups (Noether–Lefschetz theory) may be used without proof.

Exercise 0. A quartic surface

Let

$$X \subset \mathbb{P}^3$$

be a very general smooth quartic surface.

See [Huy, Chapter 1, §1, Example 1.3(i); Chapter 6, §2].

Exercise 1. A K3 surface of Picard number two

Let X be a projective K3 surface with

$$\mathrm{NS}(X) \simeq \begin{pmatrix} 2 & 0 \\ 0 & -6 \end{pmatrix}.$$

Such a surface exists by Morrison's lattice-theoretic existence theorem.
Such a surface can also be realized as a very general complete intersection of type $(1,1)$ and $(2,2)$ in $\mathbb{P}^2 \times \mathbb{P}^2$.

See [Mor, Corollary 2.9], [Weh, §0.2] and [Bar].

Exercise 2. A double cover of $\mathbb{P}^1 \times \mathbb{P}^1$

Let

$$\pi : X \longrightarrow \mathbb{P}^1 \times \mathbb{P}^1$$

be the double cover branched along a very general smooth curve

$$B \in |\mathcal{O}(4,4)|.$$

Denote by F_1, F_2 the pullbacks of the two ruling classes.

See [FM, §5.1].

Exercise 3. An elliptic K3 surface with a section

Let

$$f : X \longrightarrow \mathbb{P}^1$$

be the relatively minimal smooth elliptic surface obtained from a very general Weierstrass model

$$y^2 z = x^3 + A(t)xz^2 + B(t)z^3,$$

where

$$A \in H^0(\mathbb{P}^1, \mathcal{O}(8)), \qquad B \in H^0(\mathbb{P}^1, \mathcal{O}(12)).$$

Assume that the discriminant has 24 distinct simple zeros.
Let F be the fibre class and S the zero section.

See [Huy, Chapter 11, §2, Remark 2.1; §3, Corollary 3.4].

Exercise 4. Resolution of a nodal quartic

Let

$$Y \subset \mathbb{P}^3$$

be a very general quartic surface with exactly one ordinary double point and no other singularities.
Let

$$\pi : X \longrightarrow Y$$

be its minimal resolution.
Denote by $H = \pi^* \mathcal{O}_Y(1)$ the pullback of the hyperplane class and by E the exceptional curve.

See [FNP, §4].

Exercise 5. A lattice-polarized K3 surface

Consider the even lattice

$$L = \langle 12 \rangle \oplus \langle -4 \rangle \oplus \langle -4 \rangle$$

with basis h, e_1, e_2.
Choose a primitive embedding

$$L \hookrightarrow \Lambda_{K3} = U^{\oplus 3} \oplus E_8(-1)^{\oplus 2}.$$

Using the surjectivity of the period map, take a very general L-polarized K3 surface X with

$$\mathrm{NS}(X) \simeq L,$$

where h is chosen to be ample.

See [Mor, Corollary 2.9].

Exercise 6. A $(2,2,2)$ hypersurface

Let

$$X \subset \mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$$

be a very general smooth hypersurface of multidegree $(2,2,2)$.
Denote by H_1, H_2, H_3 the restrictions of the hyperplane classes from the three factors.

See [BvL].

Exercise 7. An elliptic K3 surface with a reducible fibre

Let

$$f : X \longrightarrow \mathbb{P}^1$$

be a very general relatively minimal elliptic K3 surface with a zero section S, having exactly one singular fibre of Kodaira type I_2 and 22 fibres of type I_1.
Write the reducible fibre as

$$F_0 = C_1 + C_2,$$

where C_1, C_2 are its irreducible components, numbered so that

$$S \cdot C_1 = 1, \qquad S \cdot C_2 = 0.$$

One may alternatively consider a fibre of type III, together with 21 fibres of type I_1.

See [Huy, Chapter 8, §3, Remark 3.9; Chapter 11].

References

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